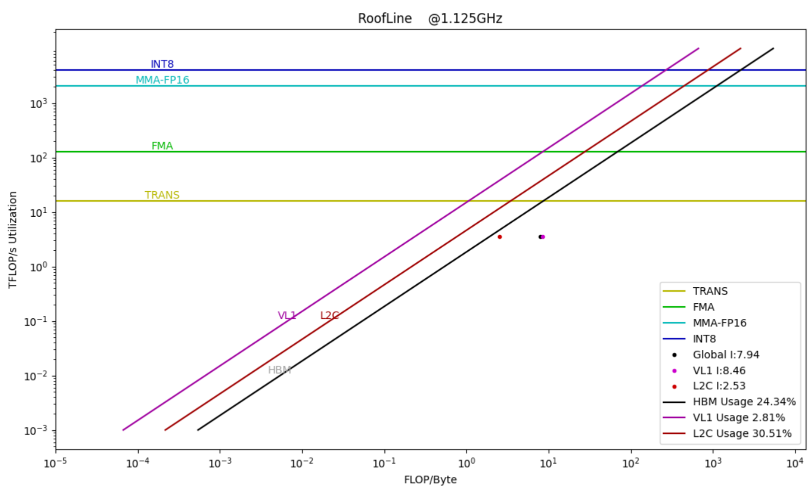
## Sub-module: Summary

| Name | Description | Value |
| --- | --- | --- |
| Total Instructions | number of all instructions | 3,583,620,439 |
| Compute Instructions | number of compute instructions | 3,245,876,521 |
| Memory Instructions | number of memory instructions | 337,743,918 |
| Total Cycles | cycles use by kernel | 9,797,435.48(Kcycles) |
| Memory Access per Second | memory access per second | 342,261.36(MB/s) |
| Average AP busy cycles | Average busy cycles per AP | 79,313,653.0 |
| AP busy Duty | average AP busy duty of total cycles | 0.81% |
| Compute Instructions busy Duty | Compute Instructions busy duty of total AP cycles | 17.64% |

Name: RoofLine

Description:




## Sub-module: CE Statistics

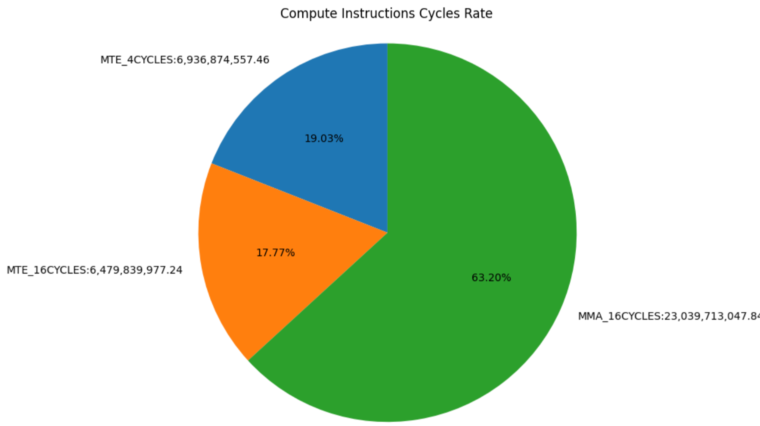
| Name | Description | Value |
| --- | --- | --- |
| WORKGROUPS | ce send workgroup | 3,028,990.0 |
| WAVES | ce send waves | 6,453,942.0 |

| | | |
|---|---|---|
| Average Wave life cycles | average life cycles per wave from start to end | 17,659.43 |

## Sub-module: ISU Statistics

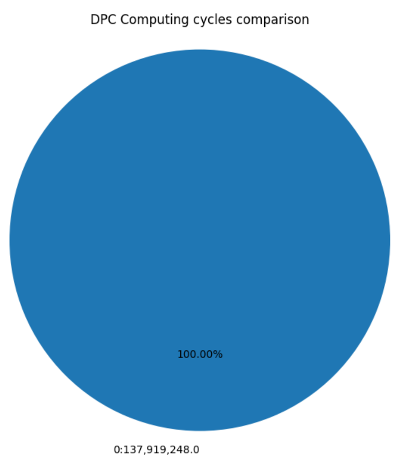
### Name: Compute Instructions Cycles Rate

Description: rate between compute instructions cycles



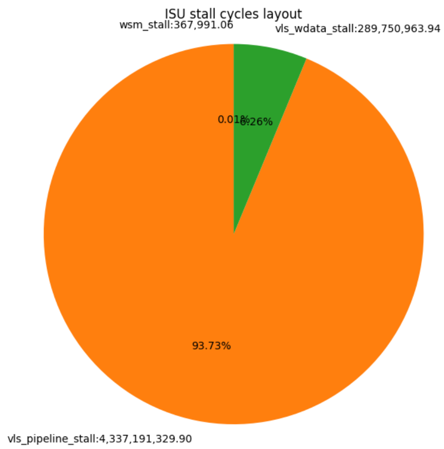

### Name: DPC Computing cycles comparison

Description: Comparison on AP computing cycles



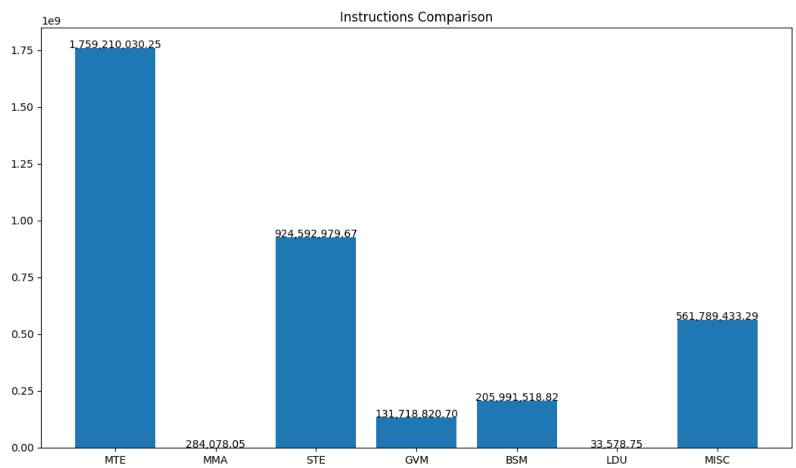
Name: ISU stall cycles layout

Description: ISU stall cycles layout




Name: Instructions Comparison

Description: Comparison between all instructions



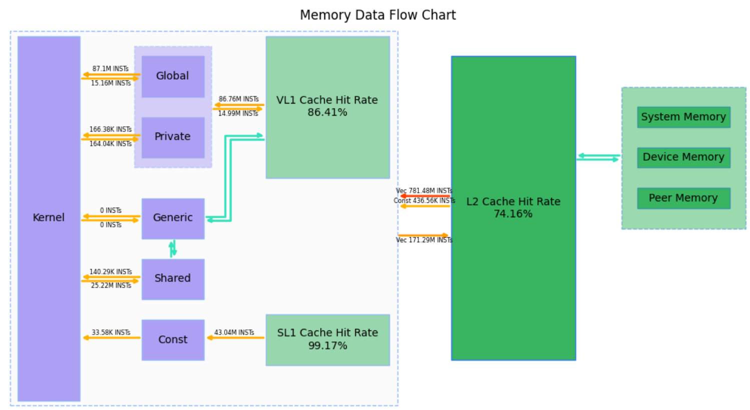

## Sub-module: Memory Statistics

| Name | Description | Value |
|---|---|---|

| | | |
|---|---|---|
| Global Read Instructions | Number of global read instructions in all APs | 87,101,571 |
| Global Write Instructions | Number of global write instructions in all APs | 15,162,708 |
| Global Atomic Instructions | Number of global atomic instructions in all APs | 29,478,849.52 |
| Private Read Instructions | Number of private read instructions in all APs | 166,378 |
| Private Write Instructions | Number of private write instructions in all APs | 164,037 |
| Constant Read Instructions | Number of smem instructions in all APs | 33,578 |
| Constant Read SL1 Instructions | Num of LDU instructions in all SL1s | 43,038,397 |
| Constant Read L2 Instructions | Num of SL1 data miss requests | 436,555 |
| Constant SL1 Hit Rate | Total LDU instructions hit rate in all SL1s | 99.17% |
| VL1 Read Instructions | total read instructions in all VL1s | 86,759,669 |
| VL1 Write Instructions | total write instructions in all VL1s | 14,989,817 |
| VL1 Atomic Instructions | total atomic instructions in all VL1s | 29,492,772.64 |
| VL1 Hit Rate | hit rate of all instructions in all VL1s | 86.41% |
| L2C Read Instructions | total read instructions in all L2Cs | 781,482,272 |
| L2C Write Instructions | total write instructions in all L2Cs | 171,289,536 |
| L2C Atomic Instructions | total atomic instructions in all L2Cs | 58,720,000.0 |
| L2C Hit Rate | hit rate of all instructions in all L2Cs | 74.16% |
| Total Atomic Instructions | total atomic instructions in memory access, include global atomic and shared memory atomic | 1,124,576.74 |
| Dnoc Read Req | number of dnoc read requests | 171,045,056.0 |
| Dnoc Write Req | number of dnoc write requests | 83,406,304.0 |
| Global Memory Read bytes | bytes read from global memory | 20,952,463,360.0byte |
| Global Memory Write bytes | bytes write from global memory | 10,674,917,376.0byte |
| Dnoc Read Average Latency | average latency of dnoc read request | 285.03 |

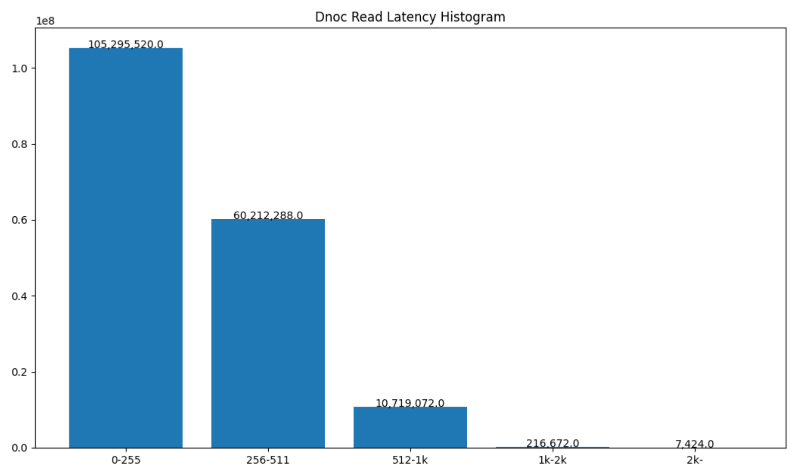
Name: Memory Data Flow Chart

Description:



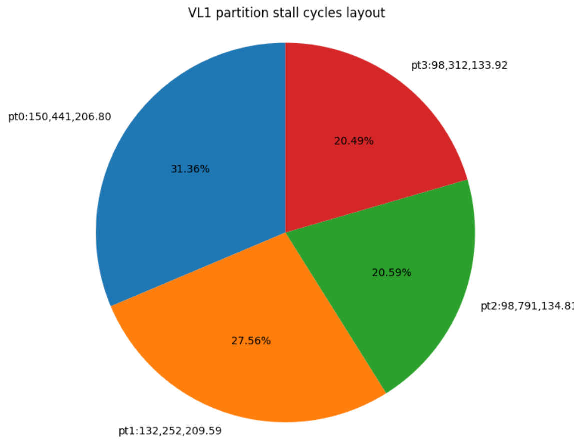

## Name: Dnoc Read Latency Histogram

Description: histogram of dnoc read latency




## Name: VL1 partition stall cycles layout

Description: VL1 partition access stall cycles layout




## Sub-module: Workgroup Memory

| Name | Description | Value |
|---|---|---|
| load instructions | number of total workgroup memory load instructions | 140,295.62 |
| store instructions | number of total workgroup memory write instructions | 25,220,545.24 |
| atomic instructions | number of total workgroup memory atomic instructions | 1,101,699.57 |
| total instructions | number of total workgroup memory instructions | 205,991,518.82 |
| average latency per load instruction | average latency per WG-load from issue to mem-ops done | 92.19 |
| average latency per store instruction | average latency per WG-store from issue to memops done | 54.44 |
| average latency per atomic instruction | average latency per WG-atom from issue to memops done | 46.60 |
| average latency of all stages per instruction | average latency of all WG-inst from issue to memops done | 99.25 |
| average cycles per load instruction | average cycles per WG-load when issuing (with bank conflict) | 10,141.04 |
| average cycles per store instruction | average cycles per WG-store when issuing (with bank conflict) | 0.01 |
| average cycles per atomic instruction | average cycles per WG-atom when issuing (with bank conflict) | 6.61 |

| | | |
|---|---|---|
| average conflict cycles per instruction | average conflict cycles per WG-inst when bank conflict happens | 3.43 |
| average cycles per instruction | average cycles per WG-inst when issuing (with bank conflict) | 6.94 |
| shared memory access efficiency | Proportion of NON-CONFLICT access | 50.55% |

## Sub-module: Occupancy

| Name | Description | Value |
|---|---|---|
| Achieved waves | number of achieved waves | 6,453,942.0 |
| Dispatched waves | number of dispatched waves | 6,454,056.0 |

## Sub-module: GPU Throughput Statistics

| Name | Description | Value |
|---|---|---|
| AP active cycles | number of wave instruction cycles executed on the AP | 8,223,063,715.18 |
| AP MTE Duty ratio | MTE Duty ratio relative to AP active | 22.33% |
| AP STE Duty ratio | STE Duty ratio relative to AP active | 11.24% |
| AP MMA Duty ratio | MMA Duty ratio relative to AP active | 17.62% |
| VLS Duty ratio | VLS Duty ratio relative to AP active | 0.0% |
| L2C Duty ratio | L2C Duty ratio relative to L2C active | 0.32% |

## Sub-module: Compute workload

| Name | Description | Value |
|---|---|---|
| instruction per cycle | The number of instructions per busy cycle across all sub-unit instances | 45.18 |
| instruction throughput | The percent of achieved instructions and theoretical instructions during all busy cycles | 0.43 |
| instruction throughput efficiency | The percentage of the maximum instruction throughput achieved during all busy cycles | 10.86% |

## Sub-module: Instruction Statistics

| Name | Description | Value |
| --- | --- | --- |
| instruction number | The total number of instructions | 3,583,620,439.54 |
| instructions per AP | The number of instructions per PEU | 34,457,888.84 |